

Worksheet 2: The Virahanka (Fibonacci) Sequence and Nature

Topic Focus: Adding previous terms the golden ratio connection and natural patterns.

1. **The Virahanka sequence is 1, 1, 2, 3, 5, 8, 13, 21, ... What is the 10th number?**

A) 34 B) 45 C) 55 D) 89

Ans: Add the two previous numbers. 9th is $13+21 = 34$. So 10th is $21+34 = 55$.

2. **Which of these natural patterns usually follows the Virahanka sequence?**

A) Legs on a dog B) Seeds in a sunflower C) Teeth in a human D) Stripes on a zebra

Ans: Spirals in sunflowers, pinecones, and pineapples often pack perfectly according to Virahanka numbers.

3. **Dividing a Virahanka number by the one just before it gets closer and closer to:**

A) The Golden Ratio B) Pi C) The Silver Ratio D) Euler's Number

Ans: As the sequence grows, the ratio converges on approximately 1.618 (The Golden Ratio).

4. **Sequence: 2, 2, 4, 6, 10, 16, ... It follows the Virahanka rule but starts with different numbers.**

What is next? A) 20 B) 22 C) 26
D) 32

Hint: Rule is $X = (X-1) + (X-2)$. Ans: Next is $10 + 16$.

5. **In the standard Virahanka sequence (1, 1, 2, 3, 5, 8, 13, 21, 34...), how often does an EVEN number appear?** A) Every 2nd number B) Every 3rd number C) Every 4th number D) Randomly

Hint: Odd + Odd = Even. Odd + Even = Odd. Even + Odd = Odd.

Ans: The pattern is Odd, Odd, Even... repeating every 3rd step.

6. **A flower has 5 petals. Another has 8. Another has 13. Following the Virahanka sequence, how many petals will the next have?** A) 18 B) 20 C) 21 D) 24

Hint: virahanka sequence with other numbers ans: Just add $8 + 13$.

7. **Find the missing number using the Virahanka rule: 4, 7, 11, __, 29, 47.**

A) 15 B) 18 C) 20 D) 22

ans: Add $7 + 11$. To check your work, see if your answer + 11 = 29.

8. **What is the sum of the first six numbers in the Virahanka sequence (1, 1, 2, 3, 5, 8)?**

A) 19 B) 20 C) 21 D) 22

Ans: $1+1+2+3+5+8$. Interestingly, formula: the sum of the first N terms is always the (N+2)th term minus 1.

9. **Which statement about the Virahanka sequence is False?**

A) Each number is the sum of the two preceding. B) It is found in pinecone spirals.
C) Every number in the sequence is prime. D) The sequence never ends.

Ans: 8, 21, 34 are in the sequence and are not prime numbers.

10. **If the 15th Virahanka number is 610, and the 16th is 987, how can you find the 14th number?**

A) Divide 987 by 2 B) Subtract 610 from 987 C) Add 610 and 987 D) Subtract 14th from 15th

Ans: Since $(14th) + (15th) = (16th)$, then $(14th) = (16th) - (15th)$.

11. **The Lucas Numbers follow the same rule but start with 2 and 1 (2, 1, 3, 4, 7...). What is the 7th Lucas number?**

A) 11 B) 18 C) 29 D) 47

Ans: Continue the addition: 2, 1, 3, 4, 7, 11, 18.

12. **If you draw squares with side lengths equal to the Virahanka sequence and arrange them in a spiral, the shape created is called a:**

A) Pythagoras Spiral B) Golden Spiral C) Archimedean Spiral D) Euler Spiral

Ans: This specific geometric pattern creates the famous spiral seen in nautilus shells.

13. **What is the difference between the 8th Virahanka number and the 7th Virahanka number?**

- A) The 5th number B) The 6th number C) 0 D) 1

Hint: If $\text{Term}(n) = \text{Term}(n-1) + \text{Term}(n-2)$, then $\text{Term}(n) - \text{Term}(n-1)$ must equal $\text{Term}(n-2)$.

14. **Look at every SECOND number in the sequence (1, 2, 5, 13...). The difference between these numbers (1, 3, 8...) forms what?**

- A) A sequence of prime numbers B) The alternate Virahanka numbers
C) Even numbers D) Square numbers

Ans: The differences (2-1=1, 5-2=3, 13-5=8) form The alternate Virahanka numbers

15. **Pingala, an ancient Indian mathematician, discovered this sequence while studying what?**

- A) Astronomy B) Sanskrit poetry meters C) Architecture D) Trade weights

explanation: The Virahanka sequence originally appeared in India centuries before Fibonacci in the context of long and short syllables in poetry.